CLASS TEST - 1

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MLA0204

Q1 BAYES DECISION:

Sol:

Given: $P(C) = 0.03$, sensitivity 0.90, specify 0.94, $\lambda(no|C) = 50$, λ (treat $|\neg C$) $= 2$.

① likelihoods: $P(+|\neg C) = 0.06$ and $P(-|C) = 0.10$.

② evidence: $P(+|\neg C) = 0.90(0.03) + 0.06(0.97)$
$= 0.0852$, so $P(-) = 0.9148$.

③ posteriors:

* $P(C|+) = 0.0270 / 0.0852 = 0.3169$,
so $P(\neg C|+) = 0.6831$.

* $P(C|-) = 0.003 / 0.9148 = 0.0033$.

④ conditional risks:

* After +: $R(\text{treat}) = 2(0.6831) = 1.3662$
and $R(no) = 50(0.3169) = 15.845$.

* After −: $R(\text{treat}) = 1.9934$ and $R(no) = 0.1640$.

⑤ Decisions: treat on +, do not treat on −.
Equivalently, treat when $P(C|x) > 2/52 = 0.0385$.

⑥ Bayes risk: $0.0852(1.3662) + 0.9148(0.1640)$
$= 0.2664$ per patient.
Check: $2(0.0582) + 50(0.0030) = 0.2664$.

⑦ second independent positive test:
* posterior odds $= (0.3169 / 0.6831) \times 15$
$= 6.959$

* p(c|+) = 0.8744.

(8) 0-1 rule: it treats only when p(c|x) since p(c|+) = 0.217, it never treats a. Its cost is 50(0.03) = 1.5. minimum risk therefore cuts the cost by 82.2%.

conclusion: A positive test gives only a 31.2% chance of disease because the prior is low. Treatment is still justified, since a missed case cost 25 times a false alarm. minimum risk (0.2664) is far better than minimum error (1.5), and a repeat test raises the probability to 87.4%.

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Q2

INFORMATION THEORY:-

* Marginals: p(x) = (0.32, 0.38, 0.30), p(y) = (0.68, 0.32)

* Entropies: H(x) = 1.5876, H(y) = 0.9044 bits

* Joint and conditional: H(x,y) = 2.2770, so H(y/x) = 0.6994 bits.

* Mutal information: I(x;y) = 0.9044 - 0.6994 = 0.2050 bits.

* cross entropy and KL: H(p,q) = 1.6766, so D_KL(p||q) = 0.0990 bits.

* fault given load: p(yes/High) = 0.20/0.30 = 0.667 (low 0.0625, medium 0.263).

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CONCLUSION :- Load carries useful but moderate information about faults. It over-weights low load and under weight high load, so it should be re-estimated from the data.

Q3 OLS AND RIDGE :

* OLS : $S_{xx} = 1000$ and $S_{xy} = 1210$, so $w_1 = 1.21$ and $w_0 = 11.7$.
* fit quality : $SSE = 13.90$, $SST = 12.70$, $R^2 = 0.992$
* prediction at $x = 60$: 90.3 ms : this is extra polation beyond 10-50
* Ridge ($\lambda = 500$) : $w_1 = 1.210 \times 1000 = 0.893$ (33% shrinkage), $\hat{y}(60) = 72.2$ ms, $SSE = 204.6$, $R^2 = 0.832$.

CONCLUSION :

$\lambda = 500$ over regularizes and under-fits. OLS is more reliable for these nearly linear data. use small λ chosen by cross validation and treat $x = 60$ forecast with caution.

Q4 NAIVE BAYES

* priors : 0.40 (spam) and 0.60 (legit).
* Smoothing Laplace $\alpha = 1$.

	offer	free	meeting
Spam	0.6667	0.5992	0.1190
legit	0.1538	0.3643	0.5484

* scores for (1,0) : spam = 0.1428, legit = 0.10079
So $p(\text{spam}|x) = 0.9884$.
* cost sensitive threshold : 5/6 = 0.833. since 0.9886 > 0.833, the email is spam
* test metrics : accuracy 83%, precision 81.8%, recall 75%, F1 = 78.2%
* cost : 5(10) + 15 = 65 units.

Conclusion :- the email is confidently spam. the independence assumption is violated because "offer" and "free" are correlated, so the posterior is over confident, throught the ranking is useful. raising the threshold or using a discriminative model would reduce false positives.

BACK - PROPAGATION:-

* Forward pass : $z_1 = 0.100$ and $z_2 = 0.550$
so, $h = (0.5250, 0.6341)$.
then $z_0 = 0.2813$, $y = 0.5650$, and $E = 0.0748$
* Deltas : $\delta_0 = -0.1069$, $\delta_1 = -0.0160$, $\delta_2 = +0.0099$.
* updated output layer : $v = (0.6281, 0.2661)$, $b_2 = 0.2535$.
* updated hidden layer : $w_{11} = 0.4040$, $w_{21} = -0.1910$, $b_1 = 0.1080$, $w_{12} = 0.2915$, $w_{22} = 0.4930$, $b_2 = 0.1050$.
* second pass : $y = 0.5378$, $E = 0.0850$ (about 10.2% lower).

conclusion :- The step moves the output toward the target and reduce the error. The hidden deltas are much smaller than the output delta because the sigmoid derivative is at most 0.25, which shows the vanishing - gradient tendency.